

TABLE IX
CONFIGURATION OF TRAINING HYPERPARAMETERS USED IN FIGURE 1 (FOLLOWING [9], LLAMA1-7B_{9k} DENOTES LLAMA1-7B LANGUAGE MODEL FINETUNED ON 9K SUBSET OF ALPACA 52K WITH THRESHOLD $\tau = 4.5$).

Model	LR	Epoch	Batch Size	Cutoff Length	Template	Scheduler	Warmup Ratio	Weight
LLaMA1-7B _{9k}	2e-5	3	128	2048	vicuna	cosine	0.1	FP32
LLaMA1-7B _{39k}	2e-5	3	128	2048	vicuna	cosine	0.1	FP32
LLaMA1-7B _{52k}	2e-5	3	128	2048	vicuna	cosine	0.1	FP32

TABLE X
CONFIGURATION OF TRAINING HYPERPARAMETERS (FOLLOWING [10, 6]).

Model	LR	Epoch	Batch Size	Cutoff Length	Template	Scheduler	Warmup Ratio	Weight
$\phi_{quality}$	2e-5	3	5	2048	-	linear	0	FP32
$\phi_{complexity}$	2e-5	3	5	1024	-	linear	0	FP32
LLaMA1-13B	2e-5	6	512	2048	vicuna	cosine	0.1	FP16
LLaMA2-13B	2e-5	6	512	2048	vicuna	cosine	0.1	FP16
LLaMA3.1-8B	2e-5	6	512	2048	vicuna	cosine	0.1	BF16
Qwen2.5-7B	2e-5	6	512	2048	vicuna	cosine	0.1	BF16

APPENDIX

B. Adding Constraints Instruction

A. Auto-grader Instruction

System Prompt:

We would like to request your feedback on the performance of AI assistant in response to the instruction and the given input displayed following.

Instruction: {instruction}

Input: {input}

Response: {response}

User Prompt:

Please rate according to the accuracy of the response to the instruction and the input. Each assistant receives a score on a scale of 0 to 5, where a higher score indicates higher level of the [accuracy]. Please first output a single line containing the value indicating the scores. In the subsequent line, please provide a comprehensive explanation of your evaluation, avoiding any potential bias.

Following [4], we use the instruction -A to prompt gpt-3.5-turbo-0301 to rate the Alpaca-52K dataset. Although the auto-grader instruction explicitly instructed the model to rate on a scale of 0 to 5, the GPT-3.5-turbo-0301 model produced an anomalous output for a very small number of instances. Out of the 52,002 evaluated samples, **exactly 4 samples received a score of -1** (Github). Alpaca 7B indicates that the LLaMA-7B model was trained on an unfiltered set of 52,002 samples. To ensure a fair and consistent comparison with the Alpaca-7B model, we included these 4 outliers in the aggregate performance calculation for this data point. We used $\tau = -1$ to represent this "no threshold" condition.

I want you act as a Prompt Rewriter.

Your objective is to rewrite a given prompt into a more complex version to make those famous AI systems (e.g., chatgpt and GPT4) a bit harder to handle.

But the rewritten prompt must be reasonable and must be understood and responded by humans.

Your rewriting cannot omit the non-text parts such as the table and code in #Given Prompt#. Also, please do not omit the input in #Given Prompt#.

You SHOULD complicate the given prompt using the following method:

Please add one more constraints/requirements into #Given Prompt#

You should try your best not to make the #Rewritten Prompt# become verbose, #Rewritten Prompt# can only add 10 to 20 words into #Given Prompt#.

'#Given Prompt#', '#Rewritten Prompt#', 'given prompt' and 'rewritten prompt' are not allowed to appear in #Rewritten Prompt#

#Given Prompt#:

{given_prompt}

#Created Prompt#:

C. Creating Deeper Instruction

I want you act as a Prompt Rewriter.

Your objective is to rewrite a given prompt into a more complex version to make those famous AI systems (e.g., chatgpt and GPT4) a bit harder to handle.

But the rewritten prompt must be reasonable and must be understood and responded by humans.

Your rewriting cannot omit the non-text parts such as the table and code in #Given Prompt#. Also, please do not omit the input in #Given Prompt#.

You SHOULD complicate the given prompt using the following method:

If #Given Prompt# contains inquiries about certain issues, the depth and breadth of the inquiry can be increased.

You should try your best not to make the #Rewritten Prompt# become verbose, #Rewritten Prompt# can only add 10 to 20 words into #Given Prompt#.

'#Given Prompt#', '#Rewritten Prompt#', 'given prompt' and 'rewritten prompt' are not allowed to appear in #Rewritten Prompt#

#Given Prompt#:

{given_prompt}

#Created Prompt#:

E. Increase Reasoning Instruction

I want you act as a Prompt Rewriter.

Your objective is to rewrite a given prompt into a more complex version to make those famous AI systems (e.g., chatgpt and GPT4) a bit harder to handle.

But the rewritten prompt must be reasonable and must be understood and responded by humans.

Your rewriting cannot omit the non-text parts such as the table and code in #Given Prompt#. Also, please do not omit the input in #Given Prompt#.

You SHOULD complicate the given prompt using the following method:

If #Given Prompt# can be solved with just a few simple thinking processes, you can rewrite it to explicitly request multiple-step reasoning.

You should try your best not to make the #Rewritten Prompt# become verbose, #Rewritten Prompt# can only add 10 to 20 words into #Given Prompt#.

'#Given Prompt#', '#Rewritten Prompt#', 'given prompt' and 'rewritten prompt' are not allowed to appear in #Rewritten Prompt#

#Given Prompt#:

{given_prompt}

#Created Prompt#:

D. Creating Concretizing Instruction

I want you act as a Prompt Rewriter.

Your objective is to rewrite a given prompt into a more complex version to make those famous AI systems (e.g., chatgpt and GPT4) a bit harder to handle.

But the rewritten prompt must be reasonable and must be understood and responded by humans.

Your rewriting cannot omit the non-text parts such as the table and code in #Given Prompt#. Also, please do not omit the input in #Given Prompt#.

You SHOULD complicate the given prompt using the following method:

Please replace general concepts with more specific concepts.

You should try your best not to make the #Rewritten Prompt# become verbose, #Rewritten Prompt# can only add 10 to 20 words into #Given Prompt#.

'#Given Prompt#', '#Rewritten Prompt#', 'given prompt' and 'rewritten prompt' are not allowed to appear in #Rewritten Prompt#

#Given Prompt#:

{given_prompt}

#Created Prompt#:

F. Increase the Brand Instruction

I want you act as a Prompt Creator.

Your goal is to draw inspiration from the #Given Prompt# to create a brand new prompt.

This new prompt should belong to the same domain as the #Given Prompt# but be even more rare.

The LENGTH and complexity of the #Created Prompt# should be similar to that of the #Given Prompt#.

The #Created Prompt# must be reasonable and must be understood and responded by humans.

'#Given Prompt#', '#Created Prompt#', 'given prompt' and 'created prompt' are not allowed to appear in #Created Prompt#.

#Given Prompt#:

{given_prompt}

#Created Prompt#:

G. Augmenting Response with Relevance Instruction

I want you to act as a Response Rewriter.
 Your goal is to enhance the quality of the response given by an AI assistant to the #Given Prompt# through rewriting.
 But the rewritten response must be reasonable and must be understood by humans.
 Your rewriting cannot omit the non-text parts such as the table and code in #Given Prompt# and #Given Response#. Also, please do not omit the input in #Given Prompt#.
 You SHOULD enhance the quality of the response using the following method:
 Please make the Response more **relevant** to the user.
 You should try your best not to make the #Rewritten Response# become verbose, #Rewritten Response# can only add 10 to 20 words into #Given Response#. 'Given Response', '#Rewritten Response', 'given response' and 'rewritten response' are not allowed to appear in #Rewritten Response#.

#Given Prompt#:
 {given_prompt}
 #Given Response#:
 {given_response}
 #Rewritten Response#:

I. Enhancing Response with Enriching Depth Instruction

I want you to act as a Response Rewriter.
 Your goal is to enhance the quality of the response given by an AI assistant to the #Given Prompt# through rewriting.
 But the rewritten response must be reasonable and must be understood by humans.
 Your rewriting cannot omit the non-text parts such as the table and code in #Given Prompt# and #Given Response#. Also, please do not omit the input in #Given Prompt#.
 You Should enhance the quality of the response using the following method:
 Please make the Response more **in-depth**.
 You should try your best not to make the #Rewritten Response# become verbose, #Rewritten Response# can only add 10 to 20 words into #Given Response#. 'Given Response', '#Rewritten Response', 'given response' and 'rewritten response' are not allowed to appear in #Rewritten Response#.

#Given Prompt#:
 {given_prompt}
 #Given Response#:
 {given_response}
 #Rewritten Response#:

H. Enhancing Response with Helpfulness Instruction

I want you to act as a Response Rewriter.
 Your goal is to enhance the quality of the response given by an AI assistant to the #Given Prompt# through rewriting.
 But the rewritten response must be reasonable and must be understood by humans.
 Your rewriting cannot omit the non-text parts such as the table and code in #Given Prompt# and #Given Response#. Also, please do not omit the input in #Given Prompt#.
 You Should enhance the quality of the response using the following method:
 Please make the Response more **helpful** to the user.
 You should try your best not to make the #Rewritten Response# become verbose, #Rewritten Response# can only add 10 to 20 words into #Given Response#. 'Given Response', '#Rewritten Response', 'given response' and 'rewritten response' are not allowed to appear in #Rewritten Response#.

#Given Prompt#:
 {given_prompt}
 #Given Response#:
 {given_response}
 #Rewritten Response#:

J. Enhancing Response with Fostering Creativity Instruction

I want you to act as a Response Rewriter.
 Your goal is to enhance the quality of the response given by an AI assistant to the #Given Prompt# through rewriting.
 But the rewritten response must be reasonable and must be understood by humans.
 Your rewriting cannot omit the non-text parts such as the table and code in #Given Prompt# and #Given Response#. Also, please do not omit the input in #Given Prompt#.
 You Should enhance the quality of the response using the following method:
 Please **increase the creativity** of the response.
 You should try your best not to make the #Rewritten Response# become verbose, #Rewritten Response# can only add 10 to 20 words into #Given Response#. 'Given Response', '#Rewritten Response', 'given response' and 'rewritten response' are not allowed to appear in #Rewritten Response#.

#Given Prompt#:
 {given_prompt}
 #Given Response#:
 {given_response}
 #Rewritten Response#:

K. Enhancing Response with Supplying Additional Details Instruction

I want you to act as a Response Rewriter.
 Your goal is to enhance the quality of the response given by an AI assistant to the #Given Prompt# through rewriting.
 But the rewritten response must be reasonable and must be understood by humans.
 Your rewriting cannot omit the non-text parts such as the table and code in #Given Prompt# and #Given Response#. Also, please do not omit the input in #Given Prompt#.
 You Should enhance the quality of the response using the following method:
 Please **increase the detail level** of Response.
 You should try your best not to make the #Rewritten Response# become verbose, #Rewritten Response# can only add 10 to 20 words into #Given Response#. 'Given Response', '#Rewritten Response', 'given response' and 'rewritten response' are not allowed to appear in #Rewritten Response#.

#Given Prompt#:
 {given_prompt}
 #Given Response#:
 {given_response}
 #Rewritten Response#:

L. Instruction Ranking Template

You are RankGPT, an intelligent assistant that can rank instructions based on their instruction-following difficulty and complexity.
 Ranking the following instructions provided by different users according to the instruction-following difficulty and complexity.
 The following are {num} instructions, each indicated by number identifier [].
 The user's instructions are:
 {instructions_at_all_evolutionary_stage}
 You will rank the {num} instructions above based on the instruction-following difficulty and complexity.
 The instructions will be listed in descending order using identifiers, and the most relevant response should be listed first, and the output format should be [] > [] > etc, e.g., [1] > [2] > etc.
 The ranking results of the {num} responses (only identifiers) is:

TABLE XI
 STATISTIC INFORMATION OF FOUR TEST DATASETS USED IN THE WIN RATE EXPERIMENTS (FIGURE 1).

Test Dataset	Data Size	Download
Koala [7]	180	HF Link
Vicuna [25]	80	HF Link
WizardEval [8]	218	GitHub
Self-Instruct [5]	252	HF Link

Algorithm 1 Data Selection

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1: Input: The data pool  $X$ , select data size  $m$ , LLM  $\theta$ , threshold  $\tau$ , decay rate  $\gamma$ 
2: Output: The selected subset  $S^m$ 
3: Sorting  $X$  with the total score  $s_{total} = \sum_{t=0}^T (s_{comp} * s_{qual} + s_{qual} * s_{know})e^{-\gamma t}$ ;
4: Obtaining the sorted pool  $X^*$ ;
5: Initialize selected dataset  $S^m = \{x_0^*\}$ 
6: for Each sample  $x \in X^*$  do
7:   //  $d(x, S^m)$  denotes the distance between  $x$  and its nearest neighbor in  $S^m$ 
8:   if  $d(x, S^m) < \tau$  then
9:      $S^m \leftarrow S^m \cup \{x\}$ 
10:  else
11:    Continue
12:  end if
13:   $X \leftarrow X \setminus \{x\}$ 
14:  if  $|S^m|$  equals to  $m$  then
15:    Break
16:  end if
17: end for

```

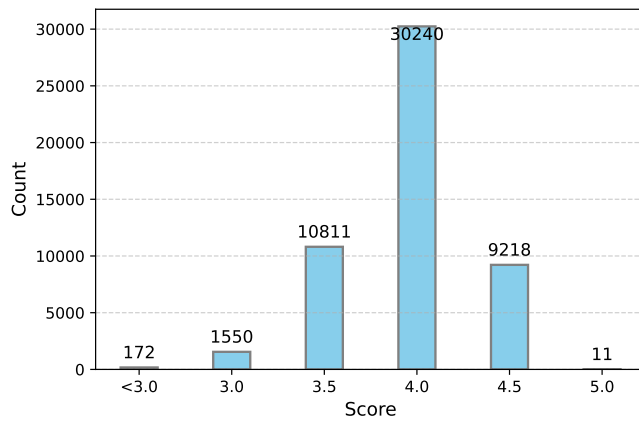
M. Response Ranking Template

You are RankGPT, an intelligent assistant that can rank responses based on their questions.
 Your evaluation should consider factors such as helpfulness, relevance, accuracy, depth, creativity, and level of detail of the response.
 The following are num responses, each indicated by number identifier [].
 You can rank them based on their relevance to the question: {given_prompt}
 {responses_at_all_evolutionary_stage}
 The user's question is:
 {given_prompt}
 You will rank the num responses above based on their relevance to the question.
 The responses will be listed in descending order using identifiers, and the most relevant response should be listed first, and the output format should be [] > [] > etc, e.g., [1] > [2] > etc.
 The ranking results of the num responses (only identifiers) is:

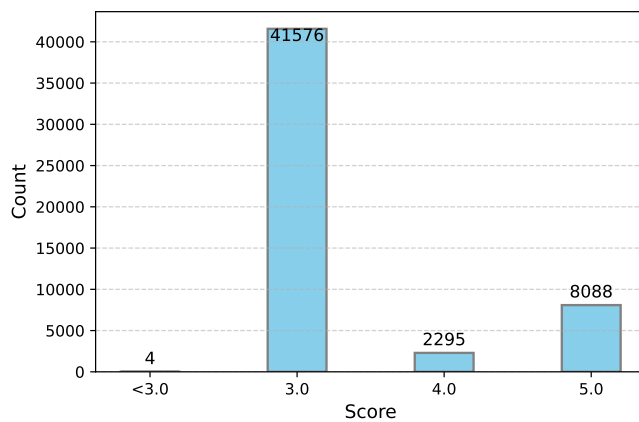
N. Data Selection Algorithm

O. Score Distribution

Following AlpaGasus [4], we prompt gpt-3.5-turbo-0301 [1] and Claude-v2 [34] to rate each data point in Alpaca dataset.



(a) Rating model: gpt-3.5-turbo-0301



(b) Rating model: Claude-v2

Fig. 8. AlpaGasus score distribution on Alpaca dataset with prompt -A. Samples with scores outside the given range were filtered out.

We report the score distribution in Figure 8. We can find that gpt-3.5-turbo-0301 and Claude-v2 tend to give similar scores. Specifically, gpt-3.5-turbo-0301 gives a 4.0 score to 58.15% of the samples, and Claude-v2 gives a score of 3.0 to 80.01% of the samples, which is difficult for subsequent data sorting. The complete set of annotated sample ratings [4], including all scores and the anomalous outputs, is available for download at: [Github](#).